Wet or Dry Soil? How Soil Water and Weed Growth Stage Affect Electrical Weed Control Efficacy in Texas Silty Clay Loam Soil

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Abstract 1498

Introduction

- **Electrical weed control (EWC)** kills weeds without herbicide or tillage, but current passes through the weed and returns to the machine **through the soil** (Slaven et al. 2023).
- Soil is an electrical conductor whose resistance depends on water content, texture and clay content:

$$\sigma_{bulk} = \sigma_w \cdot \theta \cdot T(\theta) + \sigma_s \text{ (Rhoades et al. 1976)}$$

σ_{bulk} = bulk soil conductivity; σ_w = pore-water conductivity; θ = volumetric water content; T = tortuosity (water-path connectivity); σ_s = clay-surface conductance

- **As soil dries, resistance rises**, keeping current in the root longer (Bauer et al. 2020), following Ohm's law:

$$V = I \cdot R, \text{ where } R = R_{plant} + R_{soil}$$

- However, wetter soil also conducts better and may waste less energy (Merfield 2024), so whether soil water helps or hurts EWC efficacy remains unresolved.
- **Fine-textured soils**, such as silty clay loams, which conduct along clay surfaces and crack when dry, remain untested.
- **Gap:** how soil water and weed growth stage interact to govern EWC efficacy in a silty clay loam.

Objective and Hypothesis

Objective: Determine under which soil moisture condition and at which weed growth stage EWC best controls Palmer amaranth (taproot) and Texas millet (fibrous roots) in a silty clay loam.

Hypothesis: EWC controls weeds better in dry soil than in wet soil. Dry soil resists current, so less leaks into the soil and more stays in the crown and root, giving more complete control.

Materials and Methods

Experimental design

- Texas A&M AgriLife Research Farm, near College Station, TX, 2026; silty clay loam
- Split-plot RCBD, 4 replications; main plot = soil water regime, subplot = weed growth stage
- Weeds counted in two 0.25 m² quadrats per plot, before and after EWC

2 SOIL WATER REGIMES
Irrigated (38.3% VWC)
Non-irrigated (8.7% VWC)

3 GROWTH STAGES
Early 30 · Mid 45 · Late 52 cm

2 SPECIES
Palmer amaranth (taproot)
Texas millet (fibrous roots)

WCE (%)
(Wc − Wt) / Wc × 100
c = untreated, t = treated

Soil measurements

- VWC, 0–15 cm, per plot (HydraGO, Stevens Water)
- Bulk soil EC: site mean 2 dS m⁻¹

EWC application

- Tractor-mounted Zasso, 9.1 m applicator, 6.4 km h⁻¹
- Generator: 257 V, 45 A, three-phase ≈ 34.7 kW
- Energy dose: **E/A = P / (v · w) ≈ 2.1 kJ m⁻²** (≈ 10 J plant⁻¹)

Data analysis

- Split-plot ANOVA; Fisher's protected LSD (p = 0.05)
- Python 3.11, statsmodels 0.15

Results and Discussion

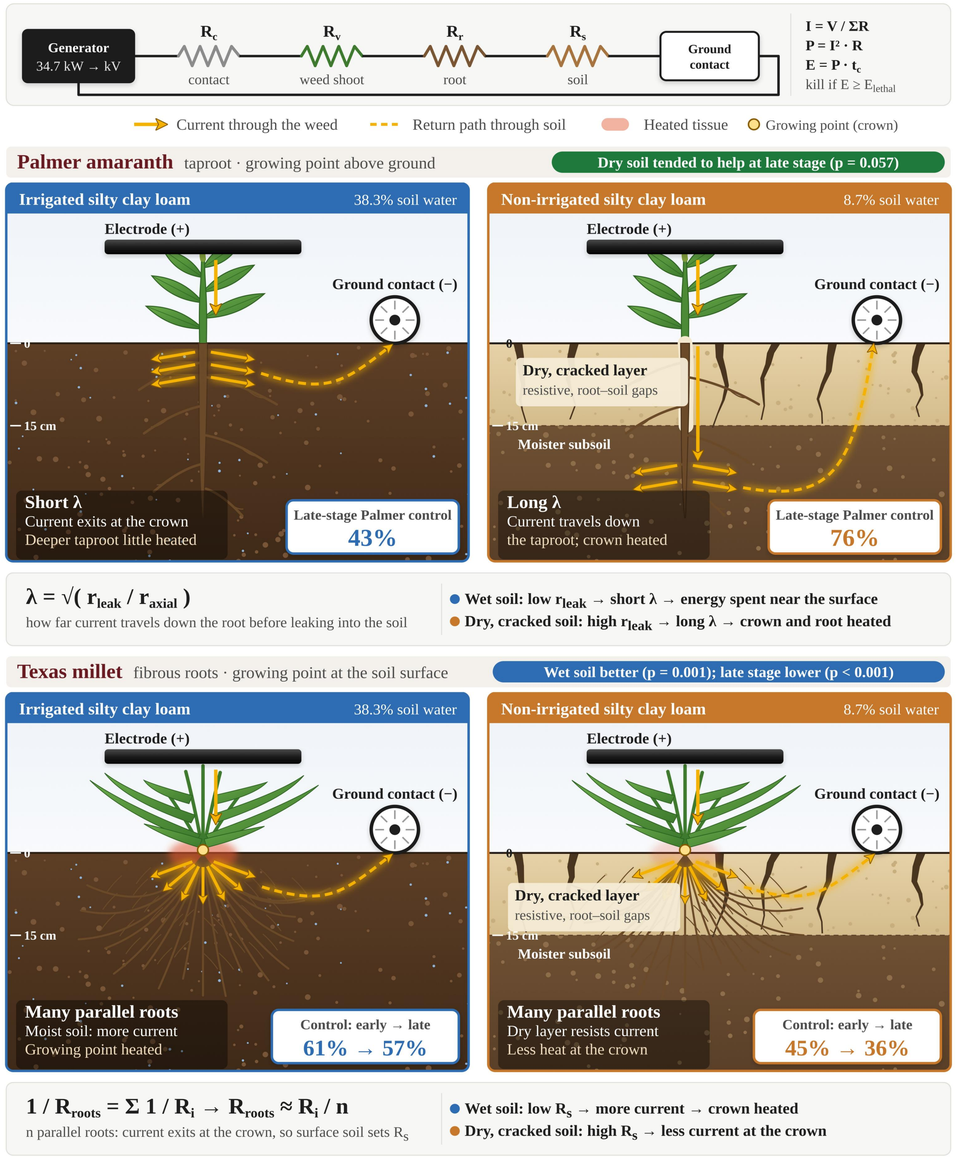


Fig 1. Conceptual current pathways in wet vs dry silty clay loam. In taprooted Palmer amaranth, dry soil lengthens λ so current heats the crown and root. In fibrous-rooted Texas millet, current exits at the crown, so moist surface soil (low R_s) delivers more current to the growing point.

- **Palmer amaranth:** late-stage WCE was 43% in wet vs 76% in dry soil (p = 0.057), consistent with Borger & Slaven (2024).
- **Texas millet:** WCE was higher in wet soil (59.5% vs 41.3%; p = 0.001) and lower at the late stage (46.3%; p < 0.001), consistent with Lati et al. (2021).
- **Surviving weeds:** at the late stage, more weeds survived EWC as soil water increased (r = 0.84, p = 0.009), consistent with Bauer et al. (2020).

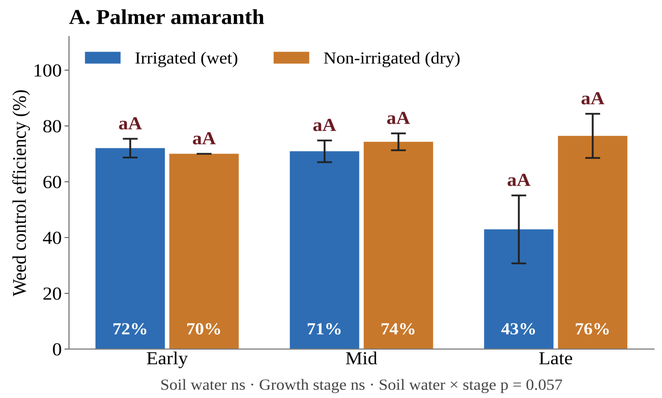


Fig 2. Palmer amaranth WCE (mean ± SE, n = 4). No significant effects; late-stage control tended to be higher in dry than wet soil (76% vs 43%; soil water × stage p = 0.057).

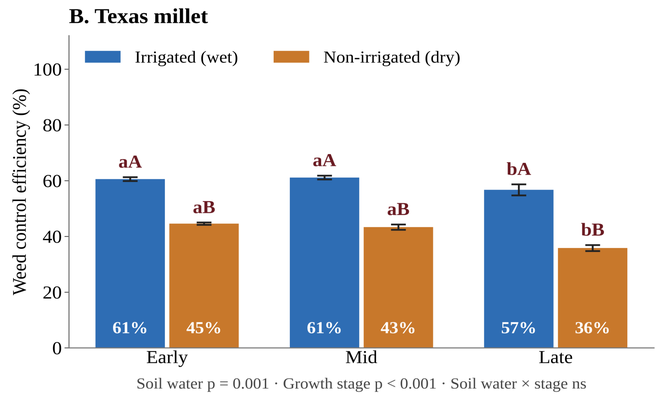


Fig 3. Texas millet WCE (mean ± SE, n = 4): higher in wet soil (p = 0.001), lower at late stage (p < 0.001). Fisher's LSD (p = 0.05): lowercase = stages, uppercase = soils.

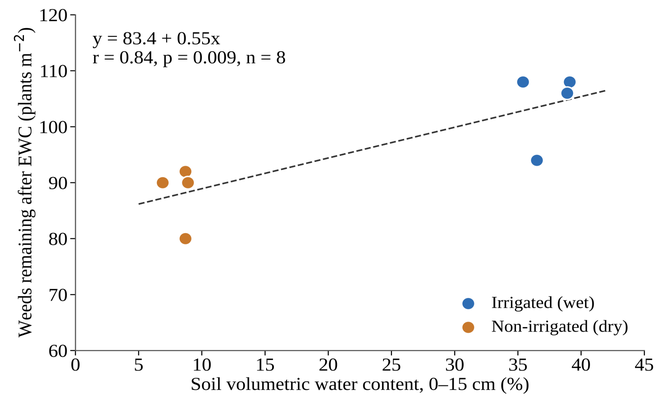


Fig 4. Late-stage weeds remaining after EWC rose with soil water (linear regression; r = 0.84, p = 0.009, n = 8). Points cluster by soil water regime.

Conclusions

- **Soil is part of the EWC circuit:** in this silty clay loam, soil water (38% vs 9%) changed weed control.
- **The hypothesis was partly supported:** soil water did not affect early- and mid-stage Palmer amaranth, but at the late stage dry soil tended to improve its control (76% vs 43%; p = 0.057). Fibrous-rooted Texas millet showed the opposite response, with better control in moist soil (59.5% vs 41.3%).
- **Soil moisture may guide EWC timing:** control grasses in moist soil; large taprooted weeds may respond better in dry soil.
- **Limits:** one site, one season; current and soil resistance were not measured.

TAKEAWAY

Time EWC by weed type and soil moisture: treat Texas millet early and in moist soil (59.5% vs 41.3%), before control drops at the late stage; late-stage Palmer amaranth tended to do better in dry soil (76% vs 43%; p = 0.057).

Future Research

- Measure soil resistivity, bulk EC, current and energy in every plot; add an intermediate soil water level and compare soil textures
- Assess regrowth at 14 and 28 days; repeat across seasons and sites

References

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